

Groupe D4: Abdelmalek Ibrahim

Autonomous Vehicle Prototype - Systems Engineering Specification

Phase 1: Project Understanding Summary

System Scope:

The system is an autonomous, electrically powered vehicle prototype. It must physically house provided off-the-shelf electronic components using 3D-printed custom mounts and a plywood chassis, strictly adhering to Additive Manufacturing constraints.

Environment & Interfaces:

The vehicle operates on a **grey floor** and must track a **2 cm wide black line** (plastic/scratch material). It is powered by an **11.5V LiPo battery**, which introduces a critical physical and electrical interface: the system must safely distribute this 11.5V power to the high-draw DC motors (via the motor controller) while ensuring the sensitive 5V logic of the Arduino is properly regulated or stepped down.

Phase 2: Refined Requirements Engineering

Updated Requirements List

1. System Requirements

- **REQ-001: Autonomous Operation** | The vehicle shall operate autonomously without human intervention during the test run.
- **REQ-002: Line Tracking** | The vehicle shall continuously detect and follow a 2 cm wide black line positioned on a grey ground surface.
- **REQ-003: Obstacle Avoidance** | The vehicle shall detect physical obstacles in its forward path and execute an avoidance maneuver.

2. Hardware & Power Requirements

- **REQ-101: Mandatory Component Usage** | The system shall exclusively use the provided electronic and structural components (1x Arduino, 1x Motor Controller, 2x DC Motors, 1x Breadboard, 1x 11.5V LiPo Battery, 2x IR-Sensors, 2x Ultrasonic Sensors, 1x Front wheel, 1x Plywood chassis).
- **REQ-102: Mechanical Fasteners** | All components shall be attached using only the provided M2x6mm and M3x8mm screws.
- **REQ-103: Power Distribution** | The system shall operate using the provided 11.5V LiPo battery, safely distributing power to both the motor controller and the microcontroller logic.

3. Manufacturing Constraints (Non-Functional)

- **REQ-201: Maximum Print Time** | The cumulative 3D printing time for all custom mounts shall not exceed 200 minutes.
- **REQ-202: Wall Thickness** | All 3D printed components shall maintain a wall thickness between 1 mm and 3 mm.
- **REQ-203: Slicing Parameters** | 3D models shall be sliced with a 0.2mm layer height and 15% infill.
- **REQ-204: Watertight Geometry** | All 3D CAD models shall be topologically closed (manifold).
- **REQ-205: Prohibited Adhesives** | The use of glue, tape, or zip ties is strictly prohibited.

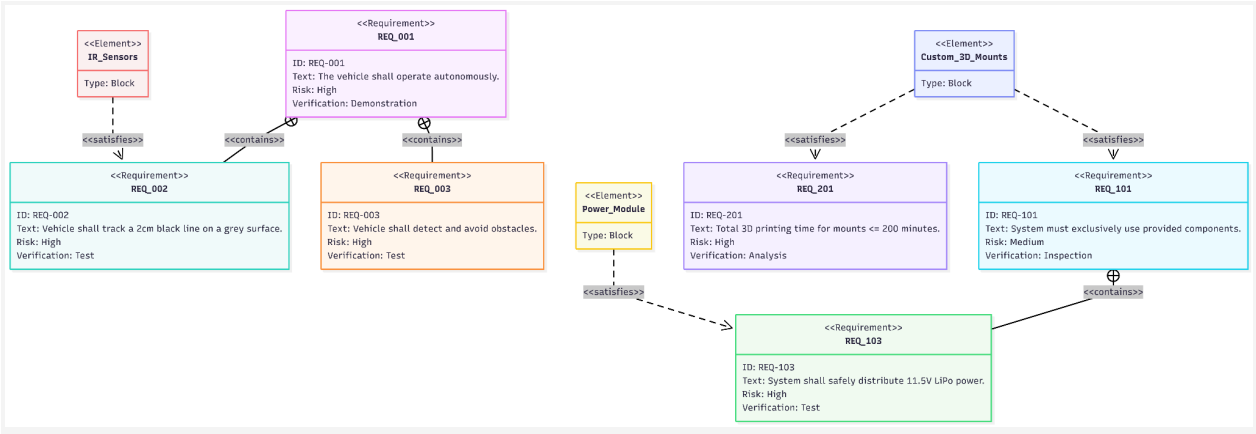
Requirements Hierarchy Table

Parent ID	Child ID	Requirement Name	Type	Relationship
-	REQ-001	Autonomous Operation	System	Root
REQ-001	REQ-002	Line Tracking (2cm Black/Grey)	Functional	Contains
REQ-001	REQ-003	Obstacle Avoidance	Functional	Contains
-	REQ-101	Mandatory Component Usage	Hardware	Root
REQ-101	REQ-102	Mechanical Fasteners	Constraint	Contains
REQ-101	REQ-103	Power Distribution (11.5V)	Interface	Contains
-	REQ-201	Maximum Print Time	Constraint	Root

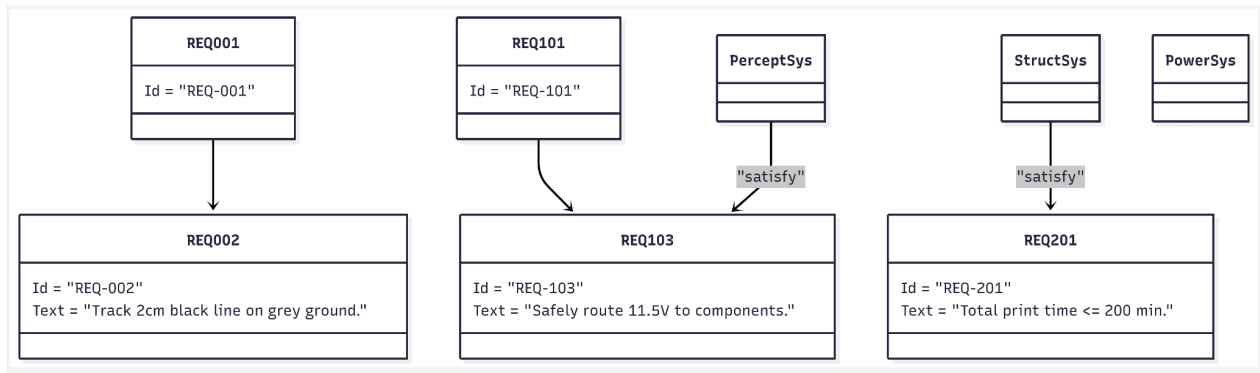
Parent ID	Child ID	Requirement Name	Type	Relationship
-	REQ-202	Wall Thickness	Constraint	Root
-	REQ-204	Watertight Geometry	Constraint	Root

Phase 3: SysML Requirements Diagram

Mermaid Diagram



PlantUML Diagram

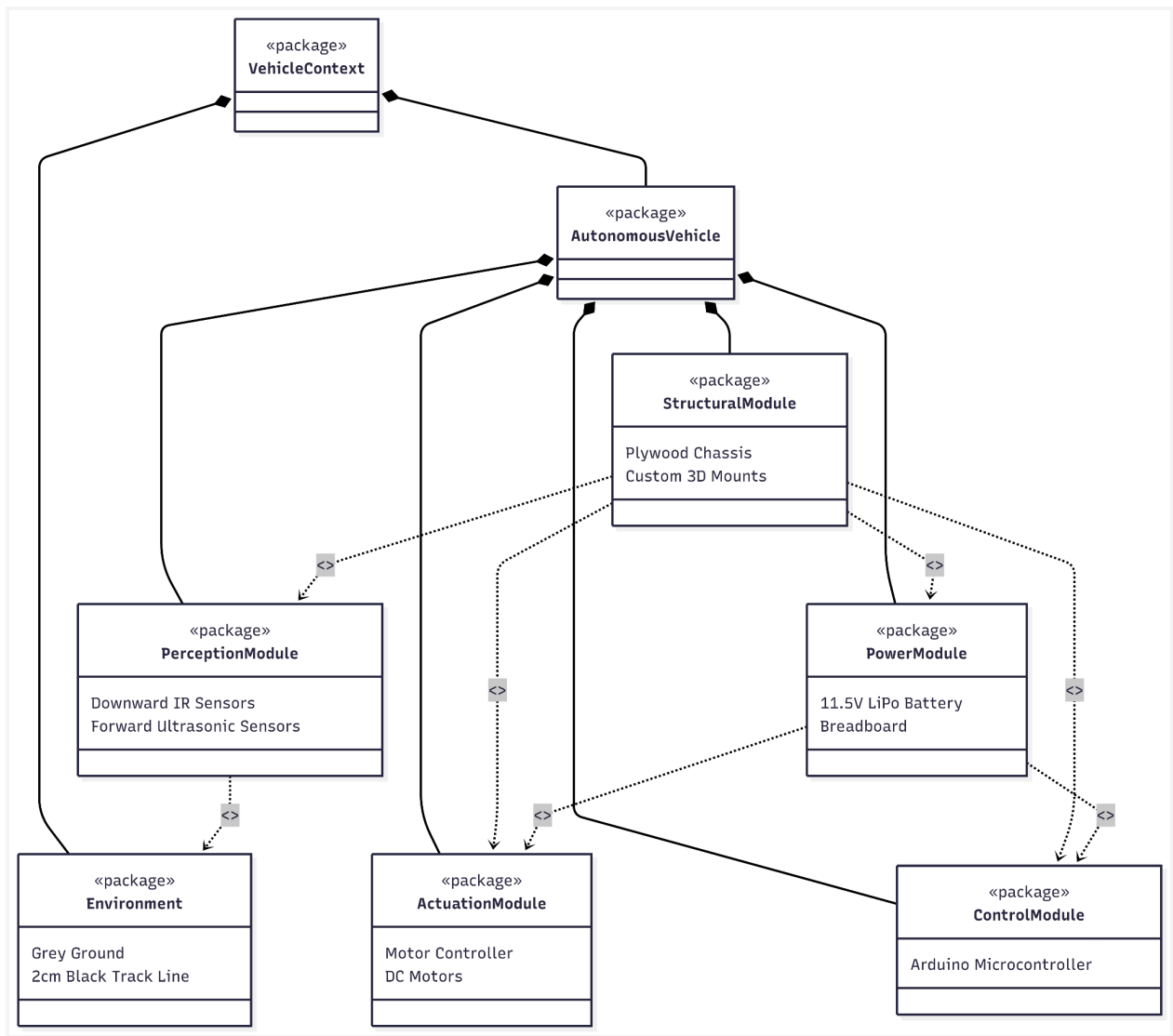


Phase 4: Structural Architecture Analysis

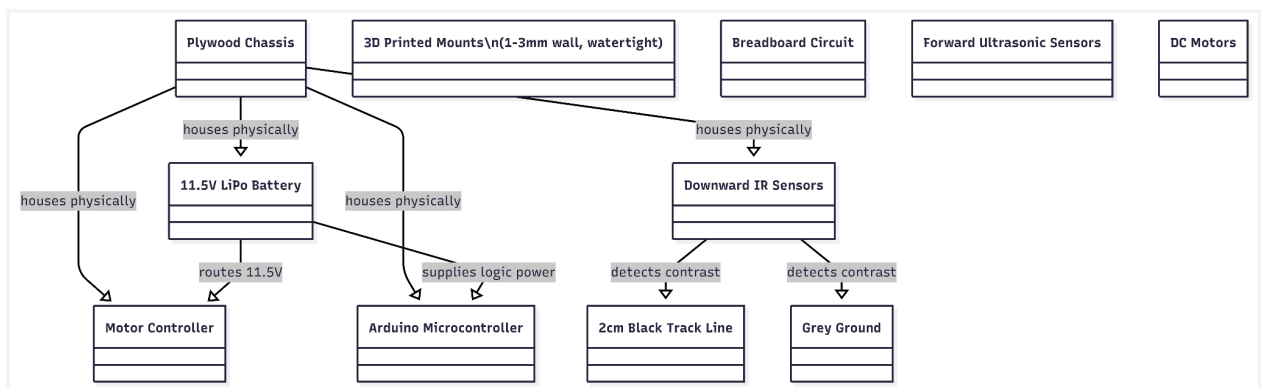
1. **Environment:** Explicitly defined by the **Grey Ground** and the **2cm Black Line**. This high-contrast physical environment dictates the placement and tuning of the IR sensors.
2. **Structural Module (Physical Foundation):** The most critical package for this phase. It acts as the physical container (<<houses>>) for all other components. It consists of the plywood chassis and the mathematically constrained 3D-printed mounts.
3. **Power Module (11.5V Distribution):** Consists of the 11.5V LiPo battery and the breadboard. It is structurally mounted via 3D parts but functionally serves to route raw power to the motor controller and stepped-down power to the logic boards.
4. **Perception & Actuation Modules:** Physically mounted to the Structural Module. The IR sensors must be mounted facing downward (to detect the 2cm line), while Ultrasonic sensors must face forward.

Phase 5: SysML Package Diagram (Physical Focus)

Mermaid Diagram



PlantUML Diagram



Phase 6: Final Validation Report

- **Completeness:** The explicit environmental constraints (grey ground, 2cm black line) and the exact power parameters (11.5V battery) are now formally documented in both the requirements and the structural architecture.
- **Methodological Accuracy:** By leaving the internal software logic abstract (ControlModule simply containing the "Arduino Microcontroller") and focusing on the StructuralModule (<<houses>> relationships) and PowerModule (<<routes 11.5V>> relationships), the models perfectly reflect your current project phase (Physical Prototyping & CAD modeling).
- **Assumptions Resolved:** No further assumptions exist. All parameters are drawn strictly from your kickoff parameters and your explicit clarifications.